

## PLINK GWAS - Data Quality Control Answers:

### Oval 1 and 2 also and Box 1 information:

```
PLINK v1.90b4.9 64-bit (13 Oct 2017)
Options in effect:
  --file GWAS
  --mind 0.10
  --out GWAS_clean_mind
  --recode

Random number seed: 1515434515
16384 MB RAM detected; reserving 8192 MB for main workspace.
Scanning .ped file... done.
Performing single-pass .bed write (6424 variants, 248 people) [Oval 1].
--file: GWAS_clean_mind-temporary.bed + GWAS_clean_mind-temporary.bim +
GWAS_clean_mind-temporary.fam written.
6424 variants loaded from .bim file.
248 people (125 males, 123 females) loaded from .fam.
1 person removed due to missing genotype data (--mind) [Box 1].
ID written to GWAS_clean_mind.irem .
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see GWAS_clean_mind.hh ); many
commands treat these as missing.
Total genotyping rate in remaining samples is 0.996863.
6424 variants and 247 people pass filters and QC [Oval 2].
Note: No phenotypes present.
--recode ped to GWAS_clean_mind.ped + GWAS_clean_mind.map ... done.
```

### Box 2a information:

```
PLINK v1.90b4.9 64-bit (13 Oct 2017)
Options in effect:
  --file MAF_greater_5
  --geno 0.05
  --out MAF_greater_5_clean
  --recode

Random number seed: 1515435189
16384 MB RAM detected; reserving 8192 MB for main workspace.
Scanning .ped file... done.
Performing single-pass .bed write (5868 variants, 247 people).
--file: MAF_greater_5_clean-temporary.bed + MAF_greater_5_clean-temporary.bim +
MAF_greater_5_clean-temporary.fam written.
5868 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see MAF_greater_5_clean.hh ); many
commands treat these as missing.
Total genotyping rate is 0.996858.
2 variants removed due to missing genotype data (--geno) [Box2a].
5866 variants and 247 people pass filters and QC.
Note: No phenotypes present.
--recode ped to MAF_greater_5_clean.ped + MAF_greater_5_clean.map ... done.
```

### Box 2b information:

```
PLINK v1.90b4.9 64-bit (13 Oct 2017)
Options in effect:
  --file MAF_less_5
  --geno 0.01
  --out MAF_less_5_clean
  --recode

Random number seed: 1515435255
16384 MB RAM detected; reserving 8192 MB for main workspace.
Scanning .ped file... done.
```

**Box 3 information:**

F1D1- Family ID for 1st individual; ID1 - Individual ID for 1st individual; F1D2- Family ID for 2nd individual; ID2 - Individual ID for 2nd individual; Z(0)- P(IBD=0); Z(1)- P(IBD=1); Z(2)- P(IBD=2); PI HAT- $P(\text{IBD}=2)+0.5*P(\text{IBD}=1)$  ( proportion IBD )

**Question 2:** How many duplicate pairs do you find (**hint:  $\text{Pi-Hat} \approx 1$** )? Do pairs with a  **$\text{Pi-Hat} \approx 1$**  have to be duplicate samples? What is another explanation? What proportion would you expect a parent/ child to share IBD? Can you find any such relationship?

There are two duplicate pairs and also a trio (two parents and a child). Parent/child relationships will have a  $\text{Pi-Hat}$  value of  $\approx 0.5$ , but so will sibpairs. We can tell that this is a parent child relationship by examine  $Z(0)$ ,  $Z(1)$  and  $Z(2)$ . We will retain only one sample from each duplicate pair and the parents NA12749 and NA12748. If you perform mixed-model analysis related individuals can be retained in the sample.

### Oval 3 information

```
PLINK v1.90b4.9 64-bit (13 Oct 2017)
Options in effect:
  --file GWAS_clean2
  --out GWAS_clean3
  --recode
  --remove IBS_excluded.txt
Random number seed: 1515440989
16384 MB RAM detected; reserving 8192 MB for main workspace.
Scanning .ped file... done.
Performing single-pass .bed write (6363 variants, 247 people).
--file: GWAS_clean3-temporary.bed + GWAS_clean3-temporary.bim +
GWAS_clean3-temporary.fam written.
6363 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
--remove: 244 people remaining.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see GWAS_clean3.hh ); many commands
treat these as missing.
Total genotyping rate in remaining samples is 0.997225.
6363 variants and 244 people pass filters and QC [Oval 3].
Note: No phenotypes present.
--recode ped to GWAS_clean3.ped + GWAS_clean3.map ... done.
```

**Table 3: Hardy Weinberg Equilibrium**

| Fail Cases |        | Fail Controls |            |
|------------|--------|---------------|------------|
| SNP        | pvalue | SNP           | pvalue     |
| None       |        | rs2968487     | 2.262e-007 |

```
PLINK v1.90b4.9 64-bit (13 Oct 2017)
Options in effect:
  --exclude HWE_out.txt
  --file GWAS_clean3
  --out GWAS_clean4
  --recode

Random number seed: 1515442367
16384 MB RAM detected; reserving 8192 MB for main workspace.
Scanning .ped file... done.
Performing single-pass .bed write (6363 variants, 244 people).
--file: GWAS_clean4-temporary.bed + GWAS_clean4-temporary.bim +
GWAS_clean4-temporary.fam written.
6363 variants loaded from .bim file.
244 people (123 males, 121 females) loaded from .fam.
--exclude: 6362 variants remaining.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see GWAS_clean4.hh ); many commands
treat these as missing.
Total genotyping rate is 0.997229.
6362 variants and 244 people pass filters and QC [Oval 4].
Note: No phenotypes present.
--recode ped to GWAS_clean4.ped + GWAS_clean4.map ... done
```

